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Sustainable alternative to bisphenol A epoxy resin: high-performance recyclable epoxy vitrimers derived from protocatechuic acid†

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Vitrimers have constituted one of the hottest research fields in recent years since they are not only mechanically robust and thermally stable like traditional thermosets but also can be recycled through a physical hot press like thermoplastics. Epoxy vitrimers are the most promising of all vitrimers because they are easy to prepare and durable for long-term service. Aromatic biomass has recently been shown to replace commercial bisphenol A as a raw material for epoxy vitrimers, but it is usually difficult to use it to achieve high performance and reprocessability at the same time. Starting from protocatechuic acid, a series of high-performance epoxy vitrimers with different catalyst loadings and ratios of epoxy/anhydride were prepared. In this work, the new epoxy resins showed much better thermal and mechanical properties ($T_g = 157\text{ }^\circ\text{C}$, tensile strength = 65 MPa) than commercial BPA-based epoxy resins. Moreover, these bio-based epoxy vitrimers can also be reprocessed with extremely high efficiency and degraded in NaOH solution. The great properties of epoxy vitrimers derived from protocatechuic acid indicate that they can be used as renewable alternatives to BPA epoxy resins.

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Introduction

Thermosetting resins have many advantages over thermoplastic materials, in particular in terms of structural stability and rigidity, and they usually show good dimensional stability, mechanical stiffness, outstanding thermal stability, and excellent physical and chemical resistance levels.¹ However, the permanent cross-linked structures of thermosets also prevent them from being recycled like thermoplastics. That is, they can only be burned or placed in landfills once they fail to work, which leads to a waste of resources and environmental pollution.² Epoxy resins account for about 70% of the market of all thermosets, and are used to form some of the most commonly used plastics in our daily life, and include encapsulating resins, adhesives, packaging materials and high-performance composite materials.³ The annual production of epoxy resins was reported to have already reached 3 million tons with a market value of over 25 billion dollars in 2017.⁴

Notably, approximately 90% of epoxy resins are produced from bisphenol A (BPA), which has been found to be harmful to human health, especially to infants, and many countries thus have set strict laws to forbid the use of BPA in baby bottles, food packaging and medical devices.⁵ Moreover, BPA is prepared from non-renewable fossil resources, which are declining at a frightening rate and are expected to be fully used up in less than 100 years.⁶ And petroleum oil is only expected to be available at current rates of usage for about 40 years. Therefore, it is necessary and urgent to find efficient solutions to achieve sustainable development of epoxy resins.

With regard to the problem of the unrecyclability of traditional thermosets, recently Leibler and his co-workers have fortunately developed a new class of thermosetting materials, called vitrimers.⁷ Unlike traditional thermosets cross-linked by irreversible covalent bonds, these materials are cross-linked by reversible dynamic covalent bonds, which can rearrange at high temperatures.⁸ As a result, vitrimers not only have the same good thermal and mechanical stabilities as do thermosets, but also exhibit desirable reprocessability levels as also do thermoplastics, thus prolonging their service lifetime effectively.^{9,10} Over the past decade, diverse vitrimers based on different reversible reactions have been reported, with these reactions including transesterification, imine exchange, radical sulfur chemistry, olefin metathesis, vinylogous urethane exchange and borate exchange.^{11,12} Of all the kinds

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of vitrimers, the epoxy vitrimers based on transesterification have attracted the most intensive attention, because they are easier to prepare and show better thermal stability when compared to other vitrimers. Furthermore, epoxy vitrimers constitute the most promising of such materials as they can be manufactured on a commercial scale, because there are only minor differences between commercial epoxy resins cured with anhydride and epoxy vitrimers.¹³ Thus the facilities used to prepare commercial epoxy resins can be directly applied to epoxy vitrimers.

BPA epoxy vitrimers based on transesterification have been studied systematically and deeply, with these studies including theoretical calculations of their relaxation dynamics,^{14–16} studies of the effects of transesterification catalysts,^{17–19} and determinations of structure–function relationships.²⁰ Apart from vitrimers based on single *trans*-esterification mechanism, many hybrid vitrimers systems have also been reported, including imine-epoxy vitrimers,^{21–23} sulfur-epoxy exchange resins^{24–27} and siloxane-epoxy resins,^{28,29} and these hybrid vitrimers often show better reprocessability and malleability. Moreover, recently, various industrial attempts at producing vitrimer composites of BPA epoxy have proved to be successful, with these composites including polyolefin-epoxy resins,^{30,31} polybutylene terephthalate (PBT)-epoxy composites³² and polymethyl methacrylate (PMMA)-epoxy materials.³³ The results of these investigations indicated that the requirements of high performance and recycling can be met at the same time. Thus, it is very likely that BPA epoxy vitrimers will be manufactured for practical functional materials in the near future.

As for the sustainability of epoxy resins, a lot of research has found that the abundant aromatic biomass in nature can be reliable candidates for BPA.^{1,3,34,35} First, they have similar phenol structures, which allows them to be as rigid and reactive as BPA. Second, they are abundant and renewable, making them sustainable for epoxy resins. Last but not least, the process of extracting useful aromatic compounds from natural macromolecules like lignin has become increasingly convenient and inexpensive with the development of chemical and biological refinement technologies. Recently, Zhang *et al.* reported producing bio-based epoxy vitrimers from eugenol,¹³ these vitrimers show good shape memory, ability to repair themselves and recyclability. Moreover, because of the presence of flexible aliphatic chains in the vitrimers, they have T_g s lower than 60 °C, which are far from satisfactory, in particular compared to those of commercial BPA epoxy resins. Later they developed other high-performance epoxy vitrimers (T_g = 187 °C) based on a rigid tri-phenol derived from vanillin and guaiacol, but nonetheless they cannot be reprocessed like typical vitrimers.³⁶ Other bio-based epoxy vitrimers also have this problem, *i.e.*, the inability to simultaneously achieve good reprocessability and high performance.^{37–41} Therefore developing sustainable high-performance epoxy vitrimers with good recyclability would be very meaningful and valuable.

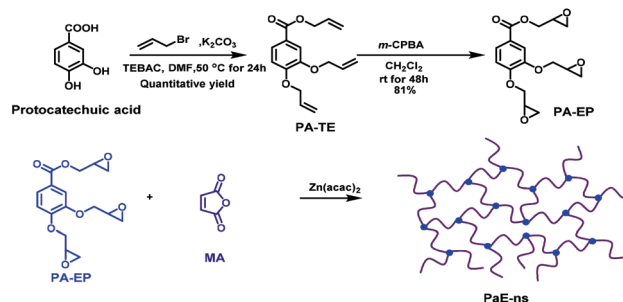
With the motivation to prepare high-performance recyclable epoxy vitrimers from renewable resources, we have made considerable efforts, and luckily found epoxy vitrimers derived

from protocatechuic acid to be able to completely meet our expectations. Protocatechuic acid is a natural tri-functional compound extracted from lignin (the second-most abundant biomass in nature) through biological fermentation.⁴² Over the past decades, much attention has been paid to its physiological and pharmacological activities.⁴³ However, to the best of our knowledge, few polymer materials have been reported based on protocatechuic acid. In our work, we prepared a series of epoxy vitrimers from protocatechuic acid and compared their properties with a commercial BPA epoxy vitrimer. These epoxy resins not only displayed much better mechanical and thermal properties than did the BPA counterpart, but were also facily recycled through a mechanical hot press with excellent recovering efficiency.

Results and discussion

Synthesis of epoxy vitrimers

We first tried the traditional two-step method to introduce epoxy groups into protocatechuic acid by reacting the acid with epichlorohydrin in the presence of phase transfer catalyst. However, no matter how we adjusted the conditions of the reaction, including reaction time, reaction temperature, and feeding amounts of epichlorohydrin and base, the yield of the target epoxy monomer **PA-EP** was very low (<30%). The poor yield was attributed firstly to many intermediates with different numbers of epoxy groups or their hydrolysed products having been produced during the epoxy functionalization process as a result of the multi-functional nature of protocatechuic acid, and secondly to the pyrocatechol structure of protocatechuic acid having led to the production of a benzo-dioxane by-product during the epoxy functionalization process and a decrease in the yield of the epoxy monomer **PA-EP**.⁴⁴ Therefore, we applied another common industrial method of epoxy functionalization, where the protocatechuic acid was first converted to its allyl ether derivative **PA-TE** and then the three allyl groups were epoxidized. The specific synthesis processes are shown in Scheme 1. It turns out that the epoxy monomer **PA-EP** derived from protocatechuic acid could be obtained in a good yield of 81%. Moreover, analyses of ¹H



Scheme 1 Syntheses of epoxy vitrimers derived from protocatechuic acid.

NMR (Fig. S1†) and ^{13}C NMR (Fig. S2†) spectra of this monomer confirmed its structure.

With the epoxy monomer **PA-EP** in hand, we tried to cure it with another bio-based succinic anhydride, but failed to prepare the expected epoxy vitrimers. Due to the very high (119 °C) melting point of succinic anhydride, the curing started before the mixed monomers formed a uniform phase, leading to incomplete curing and phase separation. However, we found a commercially available maleic anhydride (**MA**) with a melting point of as low as 55 °C, a feature making it an ideal curing agent for **PA-EP**. The processing window was allowed to be as wide as 60 °C. To thoroughly mix **PA-EP** with **MA**, we ground the mixture in a quartz mortar, and an unexpected uniform liquid with low viscosity was obtained. Then the transesterification catalyst $\text{Zn}(\text{acac})_2$ was added to this product and the resulting composition was further mixed. Notably, due to the lack of hydroxyl and carboxyl groups, which can coordinate with zinc ions in the mixed monomers, the compatibility between the monomers and catalyst was poor, thus resulting in the mixture becoming an orange batter. However, this incompatibility was overcome to some extent after curing, so decent epoxy vitrimer resins were obtained. On the other hand, it is worth mentioning that the loading of metal catalyst was designed to be less than 20% to avoid a violent curing process and phase separation issues. Table 1 shows the compositions of the different epoxy vitrimers.

The curing processes of epoxy vitrimers with different amounts of catalyst loadings were analysed by performing differential scanning calorimetry (DSC), and the results are shown in Fig. 1. As can be seen in this figure, without the transesterification catalyst $\text{Zn}(\text{acac})_2$, the curing occurred in stages and was more complex than the cases with the catalyst. This observation indicated that using this catalyst can simplify and accelerate the curing process between epoxy and anhydride with its Lewis acidity. Upon increasing the relative amount of catalyst from 5% to 10% (from **PaE-5%** to **PaE-10%**), the peak curing temperature decreased from 135 to 117 °C, illustrating that the epoxy vitrimers can be cured under milder conditions. With the motivation to prepare epoxy vitrimers under mild conditions and achieve recycling using a hot press in a relatively short amount of time, the catalyst loadings of the following epoxy vitrimers were thus all set at 10%.

Table 1 Compositions of epoxy vitrimers

Name	PA-EP/E51 (eq.)	MA (eq.)	$\text{Zn}(\text{acac})_2$ (mol%)	n (epoxy)/ n (anhydride)
PaE-0%	1.0/0	1.5	0	2/1
PaE-5%	1.0/0	1.5	5	2/1
PaE-10% or (PaE-2/1)	1.0/0	1.5	10	2/1
PaE-3/2	1.0/0	2.25	10	3/2
PaE-1/1	1.5/0	3.0	10	1/1
BADGE	0/1.0	1.0	10	2/1

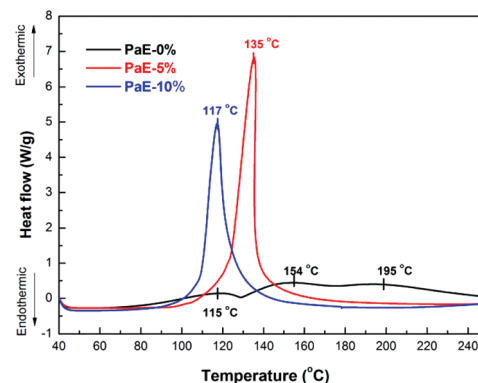


Fig. 1 DSC curves of vitrimers with different amounts of catalyst (N_2 atmosphere, 10 °C min^{-1}).

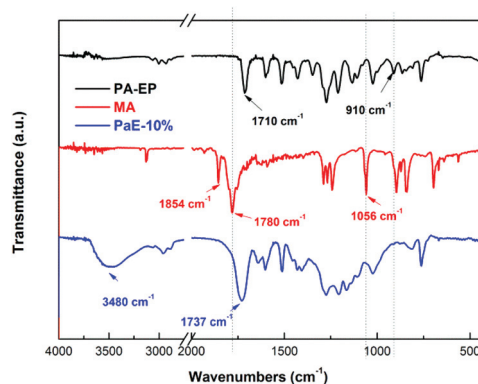


Fig. 2 FT-IR spectra of monomers and **PaE-10%**.

In order to determine whether or not the epoxy vitrimer with 10% catalyst was fully cured, FT-IR characterization of **PA-EP**, **MA** and **PaE-10%** was carried out and studied in detail. As shown in Fig. 2, for the cured resin **PaE-10%**, the typical peaks of epoxy groups of **PA-EP** at 910 and 1710 cm^{-1} were not at all observed,¹³ and neither were the characteristic anhydride peaks of **MA** at 1056, 1780 and 1854 cm^{-1} . At the same time, a new ester structure peak appeared at 1737 cm^{-1} and a new hydroxyl group peak appeared at 3480 cm^{-1} , which demonstrated that the epoxy resin **PaE-10%** had been fully cured.

It is widely acknowledged that the reason why epoxy resins can be applied to various industries is that these resins usually show good solvent resistance. Therefore, we also investigated the stability of **PaE-10%** in different solvents. The epoxy resin **PaE-10%** appeared to be an orange transparent solid, so we immersed the resin samples in different solvents for 12 h and observed their changes. As shown in Table S1,† the bio-based epoxy resin exhibited good stability in common solvents such as chloroform, toluene, ethyl acetate, *etc.*, while in other strongly polar aprotic solvents such as DMSO and DMF, the samples swelled to some degree and the colorless solvents became yellow. All in all, the epoxy resin derived from protocatechuic acid showed solvent resistance comparable to those of commercial epoxy resins.⁴

The properties of different epoxy vitrimers

In addition to catalyst loading, the proportions of epoxy monomer and anhydride can also greatly influence the performance of the epoxy vitrimers. As a result, we prepared a series of bio-based epoxy vitrimers, **PaE-ns**, with different ratios of epoxy to anhydride; and at the same time, a commercial epoxy vitrimer **BADGE** was also prepared from BPA epoxy (E51) as a control group. Their specific compositions are listed in Table 1.

DSC analyses were carried out to study the curing processes of the different vitrimers. As can be seen in Fig. 3, the onset and peak curing temperatures of the three bio-based epoxy vitrimers **PaE-ns** were all lower than those of commercial epoxy **BADGE**, indicating that these bio-based epoxy vitrimers can be cured at milder temperatures. In the **PaE-ns** system, the peak curing temperature decreased as the relative amount of **MA** was increased, and **PaE-1/1** showed the lowest peak curing temperature of 106 °C. This result was due to more **MA** providing more opportunities for the epoxy to be opened, so the epoxy resin can be cured at a lower temperature and show a lower peak curing temperature. On the other hand, when scanning for the second time, no obvious exothermic peaks were observed. Combining this observation with the complete disappearance of the epoxy and anhydride FT-IR peaks (Fig. S3†), we concluded that all of the epoxy vitrimers became fully cross-linked.

The glass transition temperature (T_g) is one of the key parameters of materials, as it determines their upper service temperature. In general, T_g can be obtained either from DSC or DMA analysis. But DSC characterization failed to present the T_g s of the epoxy vitrimers we prepared, so DMA tests (Fig. 4) were taken to obtain their T_g s, which were regarded as the peak temperatures of the $\tan\delta$ curves. The T_g s of all of the epoxy vitrimers derived from protocatechuic acid were higher than that of **BADGE**. Specifically, for the **PaE-2/1**, which has the same ratio of epoxy and anhydride, its T_g is as high as 157 °C, while **BADGE** only shows a T_g of 93 °C, indicating that bio-based epoxy vitrimers possess better thermal stabilities than commercial epoxy resins. When comparing the various

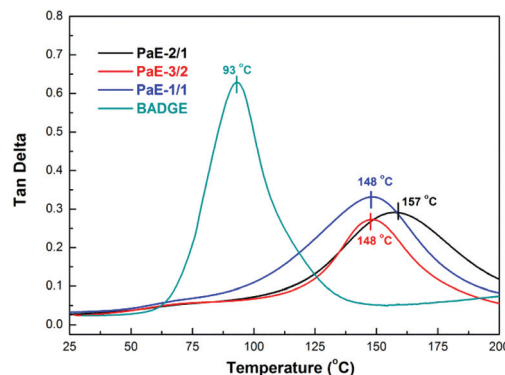


Fig. 4 DMA curves of various vitrimers (in air, 3 °C min⁻¹).

PaE-ns with different proportions of epoxy and anhydride, no considerable difference was observed between their T_g s. The T_g of a thermoset is in general controlled by both cross-link density and the rigidity of monomers. As the relative amount of **MA** was increased, although the cross-linking density may have also increased, the content of rigid epoxy monomer **PA-EP** decreased, so the T_g temperatures of **PaE-3/2** and **PaE-1/1** were lower than that of **PaE-2/1**.

In addition, the chemical thermal resistance levels of these epoxy vitrimers were also investigated by performing TGA experiments, and the results are shown in Fig. 5. The 5% weight loss temperature (T_{d5}) and amount of residual char of the epoxy vitrimers derived from protocatechuic acid were found to be much higher than those of commercial epoxy **BADGE**, indicating the better thermal resistance of the epoxy vitrimers derived from protocatechuic acid. Changing the epoxy/anhydride ratio the **PaE-ns** system did not have an obvious influence on its thermal stability, all of the **PaE-ns** samples showed T_{d5} s of about 290 °C with char yields of 33%.

In order to gain straightforward insight into the properties of the different epoxy vitrimers, their relevant performance parameters were compared and listed in Table 2. As shown in the table, for the bio-based epoxy vitrimers, the vitrimer **PaE-2/1** showed the highest T_g and T_{d5} values, the best comprehensive

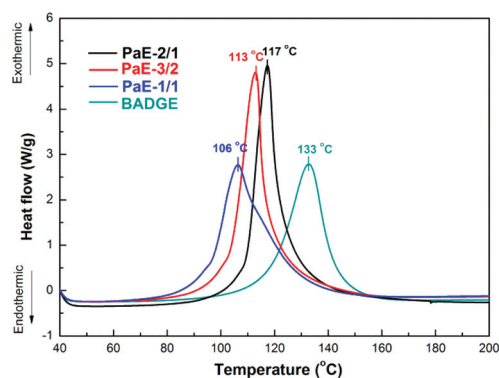


Fig. 3 DSC curves of vitrimers with different proportions of components.

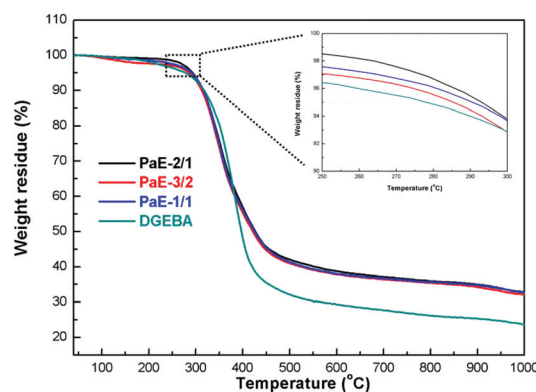


Fig. 5 TGA curves of different vitrimers in N₂ recorded at a heating rate of 10 °C min⁻¹.

Table 2 Thermal stabilities of different vitrimers

Name	T_o (°C)	T_p (°C)	T_{d5} (°C)	T_{d10} (°C)	Char (%)	T_g (°C)
PaE-2/1	82	117	293	316	33	157
PaE-3/2	73	113	288	315	33	148
PaE-1/1	73	106	290	316	33	148
BADGE	95	133	278	318	24	93

thermal properties. As for the epoxy vitrimer **BADGE** prepared from commercial bisphenol A, its onset curing temperature and thermal resistance were both found to be less desirable than those of the bio-based epoxy vitrimers. Therefore, in the following discussions, **PaE-2/1** and **BADGE** were chosen as subjects for mechanical analyses.

As can be seen from Fig. 6, although both epoxy vitrimers exhibited high tensile strength levels (>47 MPa), the bio-based vitrimer **PaE-2/1** showed much better mechanical properties, with a mechanical strength as high as 64 MPa. Moreover, its tensile modulus was measured to be as high as 2.13 GPa, 2.5 times that of **BADGE**. Based on the above results, the epoxy vitrimers derived from renewable protocatechuic acid were determined to be much better than those from commercial unrenovable BPA in terms of processability, thermal stability and mechanical properties, which indicated that protocatechuic acid can be used to replace bisphenol and achieve a sustainable production of epoxy materials.

Stress relaxation behavior of epoxy vitrimers from protocatechuic acid

From the analysis mentioned above, we determined that the bio-based epoxy vitrimers can be used as high-performance epoxy resins. Actually, as a kind of vitrimers, these materials can also be recycled through a mechanical hot press like thermoplastics due to their dynamic cross-links. Before reprocessing, it is necessary to shed light on the stress relaxation behavior in detail. Therefore, **PaE-2/1**, which was found to exhibit the best performance, was chosen for study of its stress relaxation and recycling properties.

As indicated in Fig. 7, the ratio of G/G_0 for **PaE-2/1** was found to not be equal to 0. In other words, **PaE-2/1** was shown

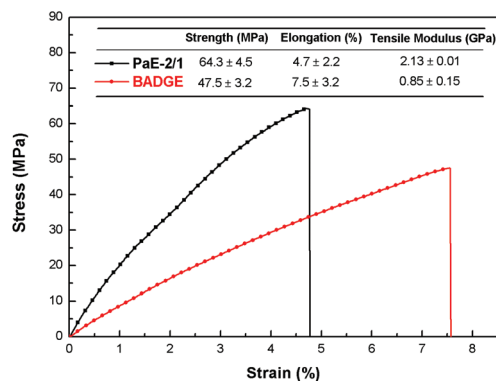
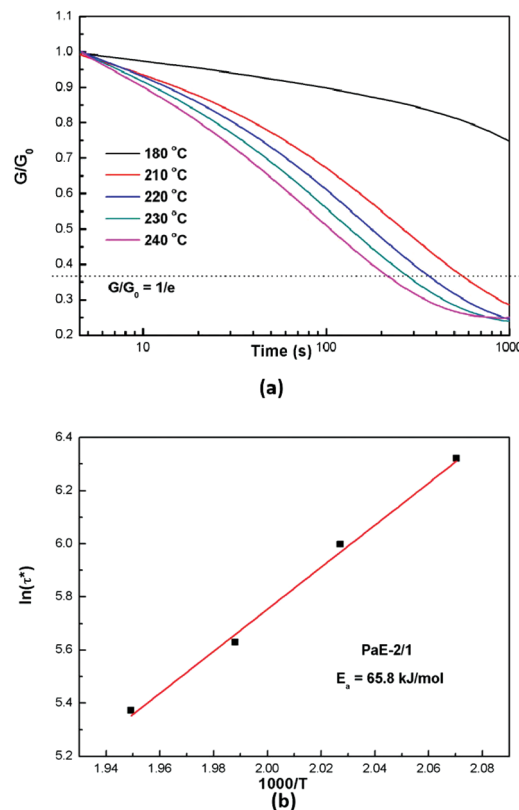
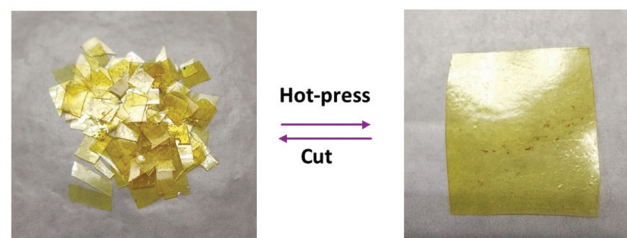
**Fig. 6** Tensile testing of **PaE-2/1** and **BADGE**.

Fig. 7 (a) Stress relaxation values of **PaE-2/1** at different temperatures. (b) The corresponding relationship between natural logarithm of relaxation time τ^* and temperature, with this relationship observed to be linear and to fit the Arrhenius law.

to be unable to fully relax at all tested temperatures, indicating a very reliable and efficient cross-linking structure of **PaE-2/1**. Due to the high T_g (157 °C) of **PaE-2/1**, the stress relaxation temperature would need to be higher than 210 °C to observe an apparent relaxation behavior. According to the characteristic relaxation time τ^* (i.e., the time needed for G/G_0 to be equal to $1/e$) obtained from Fig. 7(a), a straight line can be fitted through the data according to the Arrhenius law ($\ln \tau^* = E_a/RT + \ln A$).⁴⁵ With the slope in hand, the active energy (E_a) of the transesterification reaction catalyzed by $\text{Zn}(\text{acac})_2$ was calculated to be 65.8 kJ mol⁻¹ (Fig. 7(b)), comparable to those of many other reported epoxy vitrimers.^{7,27} The hot-press recycling of **PaE-2/1** is shown in Fig. 8.

**Fig. 8** The hot-press recycling of **PaE-2/1**.

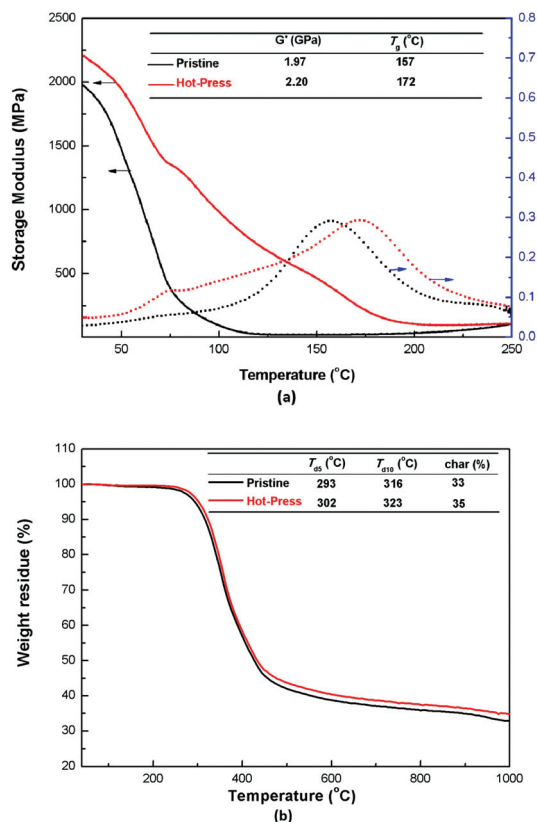


Fig. 9 Thermal stabilities of PaE-2/1 and its reprocessed samples: (a) DMA tests and (b) TGA curves.

With the motivation to study the reliability of recycling using a mechanical hot press, DMA and TGA methods were used to analyze the thermal stabilities of recycled PaE-2/1. As can be seen from Fig. 9, after being recycled through the hot press, the reprocessed PaE-2/1 samples showed both higher T_g and T_{d5} values than did the pristine PaE-2/1. Moreover, the reprocessed epoxy vitrimer also displayed a high storage modulus of up to 2.20 GPa, consistent with the tensile modulus obtained from tensile tests (Fig. S4†). The results indicated the epoxy vitrimers derived from protocatechuic acid to not only display excellent mechanical and thermal properties, but also to be amenable to being recycled effectively through mechanical reprocessing. Note in fact the better performance of the reprocessed resins than that of the original one, this result can be ascribed to the dynamic transesterification reaction having allowed the cross-linked network to recombine and rearrange to a more closely cross-linked network under hot-press conditions.

Degradability of epoxy vitrimers based on protocatechuic acid

In general, epoxy resins cured with carboxylic acids or anhydrides are usually especially susceptible to being degraded, since they are crosslinked through degradable ester bonds. For esterification reactions, changes of pH can often lead to

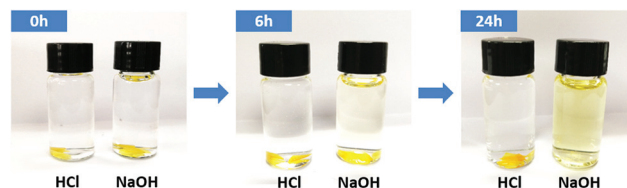


Fig. 10 Photographs of vials of PaE-2/1 in 1 M HCl (left) and 1 M NaOH (right) solutions to analyze its degradation behavior under acidic and basic conditions.

changes in equilibrium. Consequently, PaE-2/1 specimens were immersed in 1 M HCl and NaOH aqueous solution, respectively, to investigate their degradability. As shown in Fig. 10, the samples in the HCl solution were not affected after 24 hours at room temperature, while the samples in the NaOH solution degraded completely to give a transparent yellow liquid. If tested at a higher temperature (100 °C), the degradation time in the NaOH solution was observed to be shortened to 30 min. The experiments indicated the stability of the bio-based epoxy vitrimers in an acid environment but its susceptibility to being decomposed in a basic environment. This property of epoxy vitrimers may be helpful to alleviate the environmental problems involving the disposal of epoxy resins as waste.

Conclusions

Since most epoxy vitrimers are based on nonrenewable and relatively toxic bisphenol A, we developed a new kind of high-performance epoxy vitrimer from renewable protocatechuic acid, which can be obtained from fermentation of lignin. When compared with commercial BPA-based epoxy vitrimers, the bio-based epoxy vitrimers showed much better mechanical and thermal properties regardless of the epoxy/anhydride ratio. Moreover, unlike other bio-based epoxy vitrimers that do not meet the requirements of rigidness and recycling at the same time, these epoxy vitrimers can also be reprocessed through a mechanical hot press, and the recycled epoxy resins were in fact found to exhibit thermal resistance and mechanical strength levels even better than those of the original. In addition, these epoxy vitrimers can also be degraded in basic solutions. The results indicated that the renewable epoxy resins may be able to replace the commercial bisphenol A epoxy resins and find practical application in the plastics industry.

Conflicts of interest

There are no conflicts to declare.

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